AI: Strategy + Marketing (MGT 853)

AI in Retail + Fashion (Session 9)

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Yale School of Management
Spring 2026

Course Logistics

1. Guest Speakers – Please sign up for lunch (need accurate count)

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2. Project feedback ?

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3. All assignments are now complete

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8. How can we operationalize the human to AI interface?

How many Units of Items per Store per Year?

AI model with Store manager input (In class exercise)

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In class Exercise

Small Group Discussion

Forms groups of 3 or so, and explore the following questions.

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- How would you think about incentives for the manager to share this information?

Miroglio – What Happened

- After much internal debate, Francesco Cavarero and Davide Garelli decided to partner with Evo Pricing (Evo) to build an artificial intelligence (AI) system for demand forecasting and inventory replenishment for Miroglio's Elena Mirò brand.

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- Evo CEO Fabrizio Fantini and his team of data scientists worked in close collaboration with Garelli's team and with Miroglio's Chief Data Officer, to come up with the following system.

Miroglio – Designing AI system - A

Initial Inventory Allocation Decisions:

1. First decision depended on the accuracy of and confidence in the new replenishment system. Based on the system described below, Evo decided to allocate 60% (↓ from 85%) of the total inventory at the start of the season.

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5. Expensive black blouse was more likely to be allocated to stores that sold many black items (not just blouses), had higher sales of blouses (of all colors), and sold more expensive items

Miroglio – Designing AI system - B

Inventory Replenishment Decisions:

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5. Leveraging Image Data: Analyze images of fashion items to abstract key attributes (these might include not only attributes typically identified by humans, such as shape and design, but also many other unique aspects that might be hard for people to describe). It would then correlate these image attributes with sales.

Miroglio – Designing AI system - C

Including Manager Input on Customers and Local Trends:

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2. Garelli explained: "Our Head of Buying has a budget that she uses to decide how many blue jeans or black blouses to buy based on a demand forecast. However, she uses her judgment to modify these quantities from the demand forecast. We can offer a similar budget to store managers and effectively make them the buying heads of their own stores."

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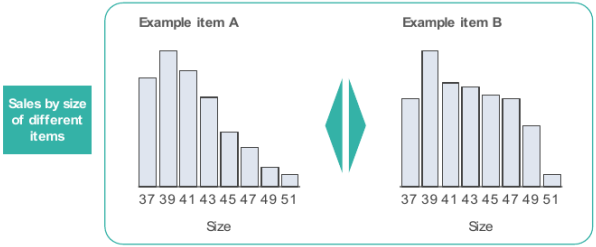
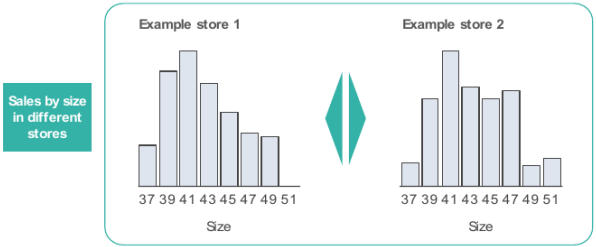
Budget

Budget = store's potential sales in the next four weeks + safety stock of 20% - current stock

Example


Sales by size of different items
Example item A
37 39 41 43 45 47 49 51
Size
Example item B
37 39 41 43 45 47 49 51
Size


Sales by size in different stores
Example store 1
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Example store 2
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Example

	Item 1	Item 2	Item 3
Selling price (€)	100	200	150
Expected sales (units)	5	3	3
Stock (units)	2	4	2

Expected revenues: €1,550 = (5 x 100) + (3 x 200) + (3 x 150)
Potential revenue from stock: €1,300 = (2 x 100) + (4 x 200) + (2 x 150)
Extra coverage: €310 = 20% x 1,550
Budget: €560 = 1,550 + 310 – 1,300

Source: Evo Pricing.

Exhibit 3 Example of Recommended Actions

Store	Item	Stock (units)	Potential (units)	Difference (units)	Action
Store 1	Item 1	10	-1	9	Mandatory release
	Item 2	1	10	-9	Urgent replenishment
Store 2	Item 1	3	1	2	Recommended release
	Item 2	6	5	1	No action
Store 3	Item 1	7	8	-1	No action
	Item 2	2	5	-3	Replenishment

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- Challenge here is that important data may NOT be available to ML system

Takeaways - 2

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Bottomline

AI challenges are embedded in a larger socio-economic context. Need to understand the ecosystem and design everything around the AI system.